



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE
Constituent of Symbiosis International (Deemed University), Pune

Department of Computer Science Engineering

Academic Year 2025-26

Batch 2023-27, Semester - VI

CA2: Design Thinking Project

Objective:

- To apply all five stages of Design Thinking (Empathy, Define, Ideate, Prototype, Test) in solving real-world problems
- To develop innovative digital solutions (Website / Mobile App)
- To encourage original thinking, creativity, and problem-solving beyond AI-generated content
- To integrate user-centric design, AI concepts, and real-world applicability
- To enhance team collaboration, research, and presentation skills

Duration: 10 days

Location: (In-Class + Outside Work)

Team: 4 students

Instructions:

- Form a team of **4 students**
- Select **ONE problem statement** from the given list
- Apply **all five stages of Design Thinking**
- Use **multiple ideation techniques** (brainstorming, mind mapping, etc.)
- Develop a **functional prototype** (Website / Mobile App – basic working or UI prototype)
- Conduct **basic user testing and feedback validation**
- Submit:
 - **Project Report**
 - **Working Link (Website/App/Prototype – mandatory)**

Important Condition:

- Direct copying from AI/tools will lead to zero marks in ideation and empathy stages
- Teams must include:
 - Hand-drawn/Original Mind Maps or Sketches
 - User Interviews / Survey Evidence
 - Iteration Proof (Version changes)
- Evaluation will focus on process over final output

Problem Statements (Choose One): More Details in the annexure**1. Wellness Assessment Platform – Website**

Develop a website for a company providing wellness assessment services integrating:

- Physical, mental, emotional wellness tracking
- AI-based insights and recommendations
- Dashboard for wellness scoring

2. Professional Portfolio Website

Develop a professional website showcasing:

- Academic profile and expertise
- Projects, research, and mentorship
- Workshops and consultancy
- Achievements and thought leadership

3. Consultancy Startup – Website

Develop a website for a consulting startup that:

- Provides industry solutions in operations and supply chain
- Includes client dashboards, diagnostics, and case studies
- Demonstrates data-driven decision-making

4. Relationship Mapping – Mobile App– Mobile App

Develop a mobile application for managing large family/organizational data integrating:

- Member profiles and detailed information
- Relationship mapping (who's who)
- Hierarchical visualization
- Search and update functionality

5. Develop a mobile application for a ride-sharing platform (similar to sRide) integrating:

- User profiles and verification
- Ride matching (route & time)
- Real-time tracking
- Ride booking and scheduling
- Cost sharing and payments
- Safety features (SOS, trip sharing)
- Ratings and feedback system

Guidelines: Design Thinking Stages

1. Empathy (User Understanding)

- Identify target users (students, professionals, industries, etc.)
- Conduct at least 5-7 user interactions/interviews
- Define user pain points

2. Define (Problem Statement)

- Frame a clear problem statement

3. Ideate (Creative Thinking)

- Use at least 2 ideation techniques: Ex. Brainstorming, Mind Mapping.etc.
- Generate multiple solution ideas before finalizing one

4. Prototype (Build Solution)

- Develop:
 - Website (HTML/CSS/JS or tools like Figma) OR
 - Mobile App (Android/iOS prototype or no-code tools)
- Include: UI/UX Design and Functional flow

5. Test (Validation)

- Take feedback from at least 5 users
- Document:
 - What worked
 - What needs improvement

Marking scheme:

S. No	Criteria	Marks
1	Empathy (User Research & Insights)	7
2	Define (Problem Statement Clarity)	3
3	Ideation (Creativity & Techniques Used)	7
4	Prototype (Functionality, UI/UX & Implementation)	10
5	Testing (User Feedback & Iteration)	3
6	Documentation with Proofs	3
7	Original Work (Authenticity)	2
8	Reality & Commercial Value	5
	Total	40

Rubrics of the evaluation component:

S. No	Component	Excellent (100%)	Average (50%)	Poor (0%)
1	Empathy	Deep insights, real users	Basic insights	No user input
2	Define	Clear & focused	Generic	Unclear
3	Ideation	Diverse & creative	Limited ideas	No ideation
4	Prototype	Functional, polished	Basic, partial	Non-functional
5	Testing	Validated & improved	Minimal feedback	No testing
6	Documentation	Complete proof	Partial proof	No proof
7	Original Work	Fully original	Partially original	Copied
8	Reality & Commercial Value	Market-ready	Somewhat practical	Unrealistic

Submission Format:

- **Team Members:** Name & PRN
- **Problem Statement Chosen:**
- **Stage-wise Work:**
 - Empathy Findings
 - Defined Problem Statement
 - Ideation Techniques & Outputs
 - Prototype Screens/Link
 - Testing Feedback
- **Final Solution Link (Mandatory):**
- **Innovations Introduced:**
- **Challenges Faced:**
- **Future Scope:**

ANNEXURE – Project Details

1. Wellness Assessment Platform – Website

To design and develop a **comprehensive website for a wellness platform** that focuses on assessing and improving human well-being using a combination of **science, Artificial Intelligence, and holistic approaches such as Ayurveda, acupressure, and energy diagnostics**.

The platform should enable users to understand their **physical, mental, and emotional wellness** through **quantified insights and personalized recommendations**.

The website must clearly communicate that the company provides **non-invasive, in-person wellness assessment services**, based on advanced scientific principles (inspired by quantum mechanics and energy-based diagnostics). These assessments are conducted physically and generate detailed wellness reports.

Additionally, the platform should highlight organizational services by offering a **Wellness Index for institutions**, such as schools, universities, corporates, and industries, to evaluate and improve their overall well-being ecosystem.

The website should be designed as a **real commercial platform**, capable of being **launched in the market**, with strong emphasis on **professionalism, credibility, and business usability**.

Key Features to Include:

- Homepage: Vision, mission, energy-based wellness concept
- About Section: Science (AI + Ayurveda + acupressure integration)
- Services Section: Individual + organizational wellness services
- Assessment Module (Simulated)
- Dashboard: Wellness scores, indices, visual reports
- Recommendations Engine: Lifestyle, nutrition, mindfulness
- Appointment Booking System
- User Profiles: Assessment history
- Blogs / Knowledge Hub

Expectation:

- Should look and function like a **real company website ready for deployment**
- Strong focus on **UI/UX, branding, and content authenticity**

2. Professional Portfolio Website

To design and develop a **professional, industry-standard portfolio website** for a Design Thinking Professor, showcasing their **academic excellence, industry experience, research contributions, and impact in innovation and entrepreneurship**.

The website should reflect the professor as a **thought leader, mentor, and innovation enabler**, highlighting their role in **teaching, mentoring startups, conducting workshops, and building innovation ecosystems**.

The platform must present a **strong personal brand**, making it suitable for:

- Academic visibility
- Industry collaborations
- Consulting opportunities
- Speaking engagements

The website should go beyond a basic profile and demonstrate **credibility, influence, and real-world impact** in the domain of Design Thinking and Innovation.

Key Features to Include:

- **Homepage:** Professional introduction, vision, and personal branding
- **About Section:** Academic qualifications, experience, philosophy of teaching Design Thinking
- **Areas of Expertise:** Design Thinking, Innovation, Entrepreneurship, AI integration.et
- **Projects & Initiatives:** Innovation cells, startup ecosystems, student projects, funded initiatives
- **Research & Publications:** Papers, patents, books, contributions
- **Workshops & Training:** Conducted sessions, seminars, corporate training
- **Mentorship & Impact:** Startups mentored, student success stories
- **Consultancy Services:** Industry collaboration, advisory roles
- **Media & Recognition:** Awards, certifications, media coverage
- **Blog / Thought Leadership:** Articles on Design Thinking, future skills, innovation
- **Contact & Collaboration:** Inquiry form, professional networking links

Expectation:

- Should reflect a **real, professional academic portfolio (ready for deployment)**
- Strong emphasis on **content authenticity and storytelling**
- High-quality **UI/UX and personal branding**
- Demonstration of **impact, not just information**

3. Consultancy Startup – Commercial Website

To design and develop a **fully functional, industry-ready website** for a startup that provides **training and consultancy services** in:

- Supply Chain Management
- Vendor Development
- Procurement & Purchase Strategy
- Six Sigma & Process Excellence
- Logistics & Distribution
- Inventory Management
- ESG (Environmental, Social, Governance)

The website should be designed as a **real commercial platform**, capable of being **launched in the market**, with strong emphasis on **professionalism, credibility, and business usability**. The platform must demonstrate how the startup helps organizations **improve efficiency, reduce costs, and achieve sustainability goals**.

Key Features to Include:

- Homepage: Value proposition & business positioning
- About Us: Vision, expertise, leadership
- Services Portfolio (detailed offerings)
- Industry Solutions (Manufacturing, FMCG, etc.)
- Case Studies / Success Stories (simulated)
- Consultation Booking / Contact System
- Client Dashboard (conceptual)
- ESG / Performance Calculator (basic logic)
- Blogs / Insights / Whitepapers

Expectation:

- Should look and function like a **real startup website ready for deployment**
- Strong focus on **UI/UX, branding, and content authenticity**

4. Relationship Mapping – Mobile App

To design and develop a **mobile application that captures, manages, and visualizes relationships within a large organization or family network.**

Consider a **case study of a family consisting of 500 members.** The application should:

- Clearly represent “**who is who**”
- Display **relationships among members**
- Store detailed **profiles and descriptions**
- Allow **dynamic updates by members**

The system should act as a **data-driven relationship intelligence platform**, useful for both families and organizations.

Key Features to Include:

- Interactive Family/Organization Tree Visualization
- Member Profiles (bio, role, details, media)
- Relationship Mapping (parent, sibling, extended links, etc.)
- Search & Filter Functionality
- Member Login & Update Capability
- Admin Control System (basic concept)
- Data Visualization (hierarchy, clusters)
- Notifications for updates/changes

Expectation:

- Clear visualization of relationships
- Scalable design for large datasets
- Focus on **data structure + UI/UX integration**
- Real-world applicability (family, corporate hierarchy, alumni networks)

5. Smart Ride Sharing & Mobility Platform – Mobile App

To design and develop a comprehensive **mobile application for a smart ride-sharing platform (similar to sRide)** that focuses on enabling efficient, cost-effective, and sustainable daily commuting using a combination of **Artificial Intelligence, location-based services, and user behavior analytics**.

The platform should enable users to **find, share, and manage rides seamlessly**, reducing travel costs, traffic congestion, and environmental impact through optimized ride matching and intelligent route planning.

The mobile app must clearly communicate that the company provides a **trusted, safe, and community-driven mobility solution**, where users can connect with verified co-travellers for daily office commutes, intercity travel, and event-based rides. The system should incorporate **real-time tracking, AI-based ride matching, and safety features** to enhance user confidence and convenience.

Additionally, the platform should highlight **organizational mobility solutions** by offering **corporate ride-sharing services**, enabling companies to manage employee transportation, reduce carbon footprint, and improve commuting efficiency through structured ride programs. The mobile application should be designed as a **real commercial product**, ready for deployment in the market, with strong emphasis on **usability, scalability, trust, and business value**.

Key Features to Include:

- **Home Screen:** Quick ride search, upcoming rides, user dashboard
- **User Registration & Verification:** Profile creation, ID verification, ratings
- **Ride Matching Engine:** AI-based matching of riders and drivers based on route, time, and preferences
- **Real-Time Tracking:** Live location tracking of rides
- **Booking & Scheduling:** Instant ride booking and future ride planning
- **Cost Sharing & Payment System:** Fare split, digital payments integration
- **Safety Features:** SOS button, trip sharing, verified users
- **Ratings & Reviews:** Feedback system for riders and drivers
- **Corporate Dashboard (Conceptual):** Employee ride management and analytics
- **Notifications & Alerts:** Ride updates, reminders, cancellations
- **Sustainability Tracker:** CO₂ savings, fuel savings insights

Expectation:

- Should look and function like a **real mobile application ready for deployment**
- Strong focus on **UI/UX, seamless navigation, and user trust**
- Demonstration of **real-world usability, scalability, and business potential**
- Integration of **smart features (AI matching, tracking, safety)**
- Clear representation of **commercial viability and market relevance**

Tip:

For more details you can connect with me for more discussion and understanding.